Introduction To Industry 4.0 And Industrial Internet Of Things

Assignment-5

Number of questions: 15 **Total marks: 15 X 1= 15**

1. What is a key technology driver in Amazon's Smart Warehousing?

a. Warehouse Automation
b. Drone Delivery
c. Cryptocurrency Payments
d. 3D Printing

Answer: (a)

Justification: Please refer to Slides 5 of Lecture 1 of Week 5

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2. Which of the following is not covered under the domain of Industry 4.0?

a. Intelligent manufacturing
b. Smart warehousing
c. Smart logistics
d. Rapid deforestation

Answer: (d)

Justification: Please refer to Slide 2 of Lecture 1 of Week 5

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3. Which of the following IIoT platforms is used by Hitachi?

a. UnifyTwin
b. Lumada
c. Insights Hub
d. RootCloud

Answer: (b)

Justification: Please refer to Slide 8 of Lecture 1 of Week 5

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4. Which of the following is not a type of business model for IoT?

a. Subscription Model
b. Outcome-Based Model
c. Asset-Sharing Model
d. Loss-Sharing Model

Answer: (d)

Justification: Please refer to Slide 12 of Lecture 2 of Week 5

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5. Which is an advantage of the Subscription Model in IoT?

a. Predictable recurring revenue
b. High initial costs
c. Limited customer interaction
d. Single revenue stream

Answer: (a)

Justification: Please refer to Slide 13 of Lecture 2 of Week 5

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6. Which is a critical challenge of the IoT-as-a-Service model?

a. Limited product applicability
b. Over-dependence on manual processes
c. Security of data
d. Reduced licensing costs

Answer: (c)

Justification: Please refer to Slide 19 of Lecture 2 of Week 5

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7. Which of the following is NOT offered by a Service-Oriented IIoT Business Model?

a. CapEx-focused investment strategies
b. Aggregation and analysis of data
c. On-demand services backed by cloud-based Business Models
d. Self-service interface

Answer: (a)

Justification: Please refer to Slide 13 of Lecture 3 of Week 5

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8. Process Oriented Business Model is a _____ business model?

a. Static
b. Actuating
c. Hybrid
d. Defunct

Answer: (c)

Justification: Please refer to Page 111 of the book Introduction to Industrial Internet of Things and Industry 4.0 by S Misra, C Roy, A Mukherjee by CRC Press

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9. What is IIRA?

a. A standard architecture for IIoT Systems
b. A scheme for corporate tax rebate
c. A package for installing IIoT Software
d. An internship program for IoT enthusiasts

Answer: (a)

Justification: Please refer to Slide 2 of Lecture 4 of Week 5

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10. Which of the following statements is incorrect about IIRA?

a. Safety is not a concern in the IIRA infrastructure,
b. The IIRA infrastructure allows interoperability
c. The IIRA infrastructure is proposed by the IIC Technology Working Group
d. The IIRA infrastructure maps application technologies

Answer: (a)

Justification: Please refer to Slide 2 and 3 of Lecture 4 of Week 5

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11. Which of the following is not a component of the three-tier architecture pattern of IIRA?

a. Edge layer
b. Fog layer
c. Platform layer
d. Enterprise layer

Answer: (b)

Justification: Please refer to Slides 15 and 16 of Lecture 4 of Week 5

12. What are the IIRA Viewpoints?

a. Mining, Smelting, Forging and Casting
b. User, Demand, Production and Supply
c. Analysis, Management, Manipulation and Modification
d. Business, Usage, Functional and Implementation

Answer: (d)

Justification: Please refer to Slide 2 of Lecture 5 of Week 5

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13. The elements of an activity in IIRA are _____.

a. Controls, Events, Feedback, and Termination
b. Cause and Effect
c. Trigger, Workflow, Effect, and Constraints
d. Planning, Projection, Execution, and Feedback

Answer: (c)

Justification: Please refer to Slide 13 of Lecture 5 of Week 5

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14. In the Business Viewpoint of IIRA, what are fundamental capabilities?

a. High-level specifications essential for business tasks
b. Detailed technical standards
c. Financial goals
d. Marketing strategies

Answer: (a)

Justification: Please refer to Slide 8 of Lecture 5 of Week 5

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15. With respect to the IIRA functional domain, industrial actuation falls under which of the following domains?

a. Operations Domain
b. Business Domain
c. Control Domain
d. Application Domain

Answer: (c)

Justification: Please refer to Slide 15 of Lecture 5 of Week 5